HHCKLA Buddhist Leung Chik Wai College

6A error analysis (2013)

Student No.: 07 Name: _____ Marks: / 60

Class: 6A Date: _____ Marked date: _____

What have you learned in this chapter? _____

What can you do to improve your work next time? _____

1. The solution of $x - \frac{x-1}{2} > 5$ or $1 < x - 11$ is

   A. $x > 9$

   B. $x > 10$

   C. $x > 11$

   D. $x > 12$

(DSE-2013 PII Q05)

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2. Let k be a constant. Solve $(x-k)^2 = 4k^2$.

A. $x = 3k$

B. $x = 5k$

C. $x = -k$ or $x = 3k$

D. $x = -3k$ or $x = 5k$

(DSE-2013 PII Q06)

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3. If a, b, c are non-zero constants and $x(x+3a)+a \equiv x^2+2(bx+c)$, then $a : b : c =$

A. 2 : 3 : 1

B. 2 : 3 : 4

C. 3 : 2 : 6

D. 6 : 4 : 3

(DSE-2013 PII Q08)

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4. Let $f(x) = x^{13} - 2x + k$, where k is a constant. If $f(x)$ is divisible by $x + 1$, find the remainder when it is divided by $x - 1$.

A. 0

B. -1

C. 2

D. -2

(DSE-2013 PII Q09)

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5. Susan sells two cars for $80 080 each. She gains 30% on one and loses 30% on the other. After the transactions, Susan

   A. loses $15 840.

   B. gains $5 544.

   C. gains $10 296.

   D. has no gain and no loss.

(DSE-2013 PII Q10)

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6. A sum of $50 000 is deposited at 8% per annum for 1 year, compounded monthly. Find the interest to the nearest dollar.

A. $4 000

B. $4 122

C. $4 143

D. $4 150

(DSE-2013 PII Q11)

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7. It is given that z varies directly as x and inversely as $\sqrt{y}$. If y is decreased by 64% and z is increased by 25%, then x

A. increases by 20%.

B. increases by 80%.

C. decreases by 25%.

D. decreases by 75%.

(DSE-2013 PII Q13)

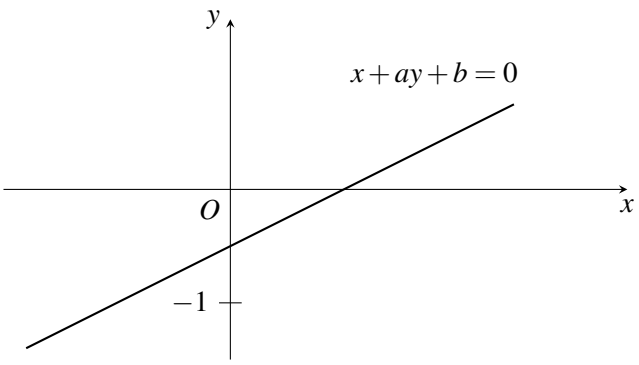
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8. The figure shows $x+ay+b=0$. Which statements are true?

I. $a<0$ II. $b<0$ III. $a<b$



A. I and II only

B. I and III only

C. II and III only

D. I, II and III

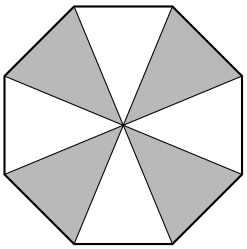
(DSE-2013 PII Q14)

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9. In the figure, the regular octagon is divided into eight identical isosceles triangles and four of them are shaded. The number of axes of reflectional symmetry of the octagon is



A. 2

B. 4

C. 8

D. 16

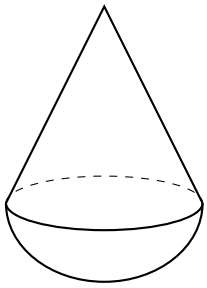
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10. In the figure, the solid consists of a right circular cone and a hemisphere with a common base. The base radius and the height of the circular cone are 3 cm and 4 cm respectively. Find the total surface area of the solid.



A. $30\pi\,\text{cm}^2$

B. $33\pi\,\text{cm}^2$

C. $48\pi\,\text{cm}^2$

D. $51\pi\,\text{cm}^2$

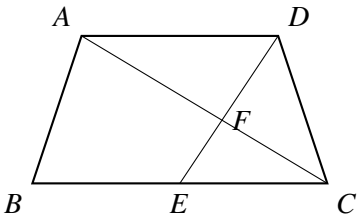
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11. In the figure, $ABCD$ is a trapezium with $AD//BC$, and $AD : BC = 2 : 3$. Let E be the midpoint of BC. AC and DE intersect at F. If the area of ΔCEF is $36\,\text{cm}^2$, then the area of the trapezium $ABCD$ is.



A. $216\,\text{cm}^2$

B. $264\,\text{cm}^2$

C. $280\,\text{cm}^2$

D. $320\,\text{cm}^2$

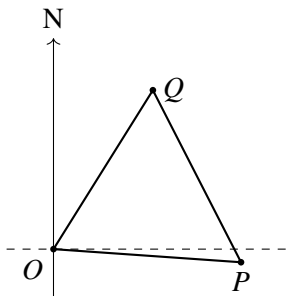
(DSE-2013 PII Q18)

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12. In the figure, the bearing of P from O is $S86^\circ E$ and the bearing of from O is $N32^\circ E$. If P and Q are equidistant from O, then the bearing of P from Q is



A. N24°W

B. N27°W

C. S24°E

D. S27°E

(DSE-2013 PII Q20)

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13. An interior angle of a regular n-gon is four times an exterior angle. Which are true?

I. $n = 10$.

II. The number of diagonals of the polygon is 10.

III. The number of folds of rotational symmetry of the polygon is 10.

A. I only

B. II only

C. I and III only

D. II and III only

(DSE-2013 PII Q21)

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14. In $\triangle ABC$, $AB : BC : AC = 8 : 15 : 17$. Find $\cos A : \cos C$.

A. 8 : 15

B. 8 : 17

C. 15 : 8

D. 15 : 17

(DSE-2013 PII Q22)

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15. The coordinates of the points A and B are $(2,5)$ and $(4,-1)$ respectively. Let P be a moving point in the rectangular coordinate plane such that $AP = BP$. Find the equation of the locus of P.

A. $x-3y+3=0$

B. $x-3y-7=0$

C. $x-3y+13=0$

D. $3x+y-11=0$

(DSE-2013 PII Q24)

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16. The equation of the circle C is $2x^2+2y^2-4x+8y-5=0$. The coordinates of the points P and Q are $(-1,2)$ and $(4,0)$ respectively. Which of the following is/are true?

I. The radius of C is 5.

II. The midpoint of PQ lies outside C.

III. If G is the centre of C, then $\angle PGQ$ is acute angle.

A. I only

B. II only

C. I and III only

D. II and III only

(DSE-2013 PII Q25)

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17. Two numbers are randomly drawn at the same time from seven cards numbered 1, 2, 3, 4, 5, 6 and 7 respectively. Find the probability that the product of the numbers drawn is an odd number.

A. $\frac{2}{7}$

B. $\frac{4}{7}$

C. $\frac{12}{49}$

D. $\frac{16}{49}$

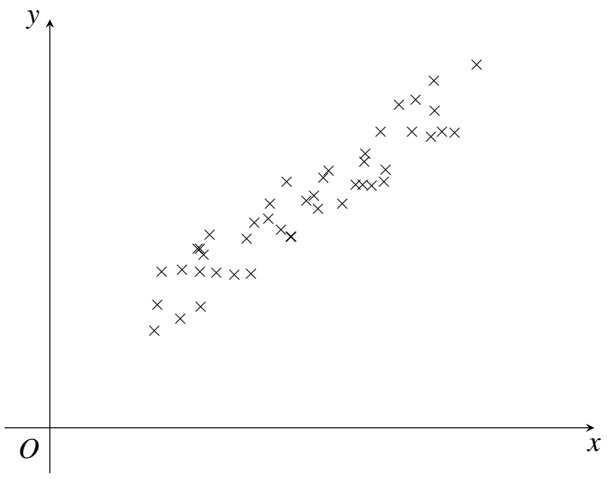
(DSE-2013 PII Q26)

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18. The scatter diagram shows the relation between x and y. Which may represent the relation?



A. y increases when x increases.

B. y decreases when x increases.

C. y varies inversely as x^2.

D. y varies directly as x^{-3}.

(DSE-2013 PII Q28)

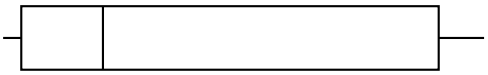
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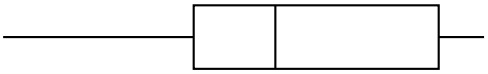
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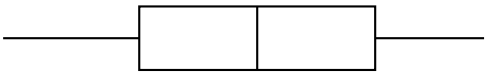
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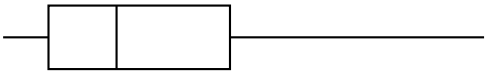
19. The stem-and-leaf diagram shows hourly wages. Which box-and-whisker diagram may represent them?

4	0 2 2 2 4 4 4 7
5	0 0 1 2 2 6 8 9
6	3 5 5 7
7	0
8	2 6
9	5

A. 

B. 

C. 

D. 

(DSE-2013 PII Q29)

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20. $A00000E00001_{16} =$

A. $10 \times 16^{11} + 14 \times 16^5 + 17$

B. $11 \times 16^{11} + 15 \times 16^5 + 17$

C. $10 \times 16^{12} + 14 \times 16^6 + 272$

D. $11 \times 16^{12} + 15 \times 16^6 + 272$

(DSE-2013 PII Q33)

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21. If $x - \log y = x^2 - \log y^2 - 10 = 2$, then $y =$

A. 100

B. 2 or -4

C. $\frac{1}{100}$ or 10000

D. $\frac{1}{10000}$ or 100

(DSE-2013 PII Q34)

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22. If $\alpha \neq \beta$ and $\begin{cases} 3\alpha = \alpha^2 - 5 \\ 3\beta = \beta^2 - 5 \end{cases}$, then $\alpha\beta =$

A. 3

B. -3

C. 5

D. -5

(DSE-2013 PII Q35)

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23. The real part of $i+2i^2+3i^3+4i^4$ is

A. 2

B. -2

C. 6

D. -6

(DSE-2013 PII Q36)

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24. Consider the following system of inequalities:

$$\begin{cases} x \geq 2 \\ y \geq 0 \\ x + 4y \leq 22 \\ 4x - y \leq 20 \end{cases}$$

Let D be the region which represents the solution of the above system of inequalities. If (x, y) is a point lying in D, then the greatest value of $3y - 4x + 15$ is

A. 3.

B. 17.

C. 22.

D. 30.

(DSE-2013 PII Q37)

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25. The nth term of a sequence is $2n - 19$. Which are true?

I. 25 is a term of the sequence..

II. The sequence has 10 negative terms.

III. The sum of the first n terms of the sequence is $n^2 - 18n$.

A. I only

B. II only

C. I and III only

D. II and III only

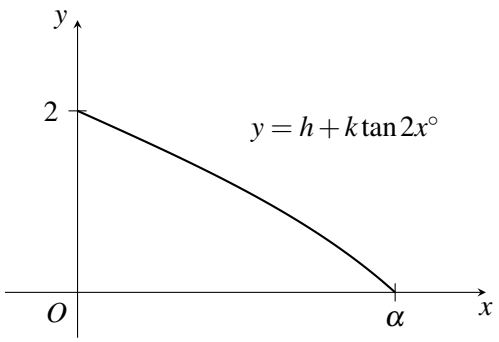
(DSE-2013 PII Q38)

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26. Let h and k be constants. The figure shows the graph of $y = h + k\tan 2x^\circ$, where $0 \le x \le \alpha$. Which of the following are true?



I. $h > 0$

II. $k < 0$

III. $\tan\alpha = \frac{1}{k}$

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

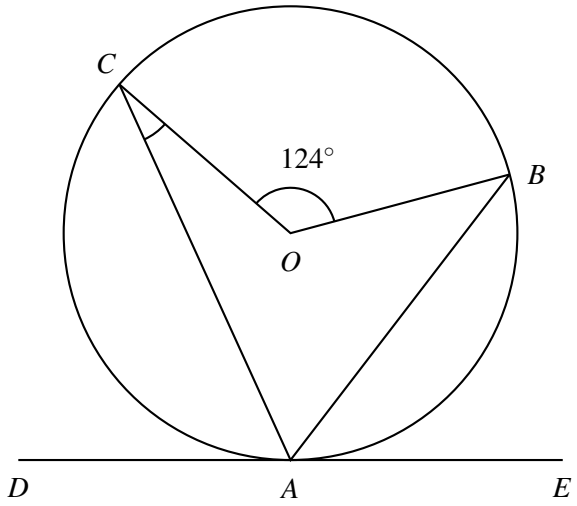
(DSE-2013 PII Q39)

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27. In the figure, O is the centre of the circle ABC, and DE is tangent to the circle at A. If AB is the angle bisector of $\angle CAE$, then $\angle ACO =$



A. 26°

B. 28°

C. 31°

D. 34°

(DSE-2013 PII Q41)

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28. Let O be the origin. If the coordinates of the points A and B are $(0, 12)$ and $(30, 12)$ respectively, then the $y-$coordinate of the circumcentre of ΔOAB is

A. 6.

B. 8.

C. 12.

D. 15.

(DSE-2013 PII Q43)

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29. If the first three digits and the last five digits of an eight-digit phone number are formed by a permutation of $5, 6, 9$ and a permutation of $2, 3, 4, 7, 8$ respectively, how many different eight-digit phone numbers can be formed?

A. 15

B. 126

C. 720

D. 40 320

(DSE-2013 PII Q44)

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30. If the variance of the five numbers x_1, x_2, x_3, x_4 and x_5 is 13, then the variance of the five numbers$3x_1 + 4, 3x_2 + 4, 3x_3 + 4, 3x_4 + 4$ and $3x_5 + 4$ is

A. 39

B. 43

C. 117

D. 121

(DSE-2013 PII Q45)

I can finish _____ questions

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